CIS 215 LAB REPORT

Network Security Fundamentals

6.2.7 Lab — Configure Automated Security Features

BACKGROUND	
Course:	CIS 215 - Network Security Fundamentals
Term:	Spring 2026
Instructor:	Veliz E.
Section Number:	14381
Student Name:	Donaven Bruce
Student ID:	900*********
Submission Platform:	Canvas / File Upload as Assigned
Report Date:	2026-04-20
Report Version:	Final
Report Title:	6.2.7 Lab - Configure Automated Security Features
Week / Module Reference:	Week 6 / Router Hardening and Automated Security
Assignment Reference:	Lab 6.2.7 / Packet Tracer Activity
Lab Type:	Hands-On Lab / Packet Tracer
Date Completed:	2026-04-20
Time Spent:	Approximately 2.0 hours

ENVIRONMENT

Platforms, Devices, and Systems Used:

- Primary host system: Windows desktop environment with Cisco Packet Tracer used for build, testing, and verification.
- Lab platform: Cisco Packet Tracer simulation with the topology built manually from a blank workspace.
- Devices used: routers R1, R2, and R3; switches S1 and S3; hosts PC-A and PC-C.
- Router platform used in the simulation: ISR4321; switch platform used: 2960-24TT.
- Security focus area: manual administrative hardening on R1 and automated hardening with AutoSecure on R3.

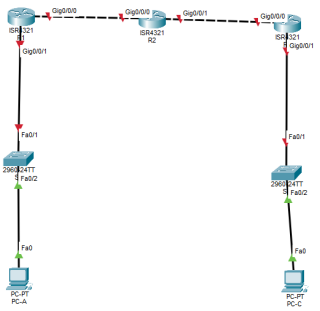


Figure 1. Initial Packet Tracer workspace after the required routers, switches, and PCs were placed and wired.

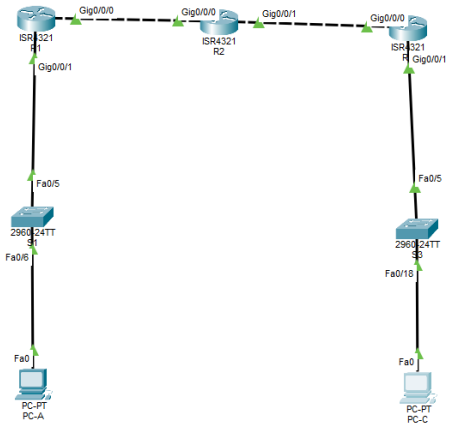


Figure 2. Final completed topology after addressing, routing, security configuration, and verification were finished.

Files, Media, and Resources Used:

- Packet Tracer source file: completed 6.2.7 Configure Automated Security Features .pkt lab file.
- Lab instructions: assignment sheet containing the topology, addressing plan, OSPF tasks, and AutoSecure workflow.
- Evidence collected: topology screenshots, OSPF verification screenshots, end-to-end ping screenshot, AutoSecure output, and final SSH verification screenshot.
- Supporting notes: per-router IP plan, OSPF checklist, credential tracking from the lab sheet, and SSH troubleshooting notes.

Work Objective:

- Build the required three-router network and cable every device according to the given topology and addressing table.
- Configure interface addressing, static host addressing, and OSPF routing, then verify end-to-end communication.
- Manually secure R1, run AutoSecure on R3, and confirm secured remote administrative access with SSH.

Learning Objectives:

- Configure interface addressing on multiple routers from a supplied addressing table.
- Deploy OSPF process 1 and verify full adjacency and learned routes across the transit topology.
- Apply passive-interface correctly on LAN-facing ports while still advertising the LAN networks into OSPF.
- Configure manual administrative security settings on one router and compare them against AutoSecure on another router.
- Verify that SSH works after the router hardening process is complete.

Scope and Constraints:

- In scope: topology build, interface addressing, PC addressing, OSPF routing, connectivity tests, R1 manual hardening, R3 AutoSecure execution, SSH verification, and final save.
- The work was completed entirely inside Cisco Packet Tracer using manual device placement, manual cabling, and router CLI work.
- The simulation used ISR4321 routers, so a few commands differed slightly from the lab sheet syntax, but the required outcomes and provided credentials were preserved.

Addressing Plan Used:

- R1 G0/0/0: 10.1.1.1 / 255.255.255.252
- R1 G0/0/1: 192.168.1.1 / 255.255.255.0
- R2 G0/0/0: 10.1.1.2 / 255.255.255.252
- R2 G0/0/1: 10.2.2.2 / 255.255.255.252
- R3 G0/0/0: 10.2.2.1 / 255.255.255.252
- R3 G0/0/1: 192.168.3.1 / 255.255.255.0
- PC-A NIC: 192.168.1.3 / 255.255.255.0 / default gateway 192.168.1.1
- PC-C NIC: 192.168.3.3 / 255.255.255.0 / default gateway 192.168.3.1

TECHNICAL WORK PERFORMED

Preparation Performed:

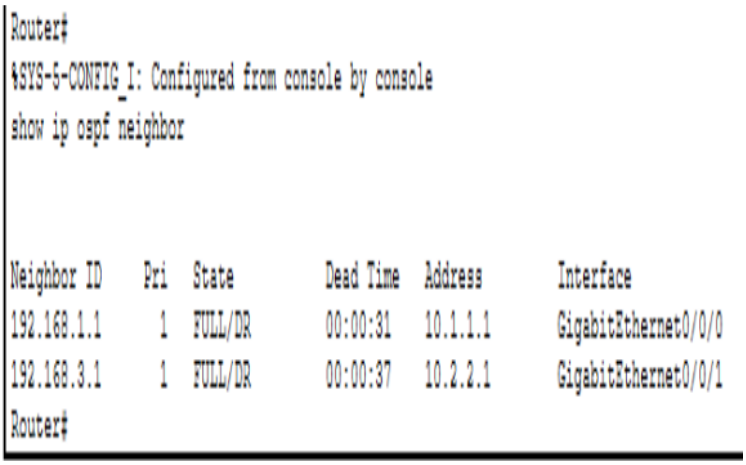
- Reviewed the topology and addressing plan before beginning the build so the interface mapping would match the lab exactly.
- Placed the routers, switches, and PCs on the Packet Tracer board, then wired each device to the specified interfaces.
- Verified the switch-port and router-port mapping before beginning CLI work so the later routing verification would align with the addressing table and screenshots.

Topology and Basic Configuration Procedure:

1) Placed and renamed R1, R2, R3, S1, S3, PC-A, and PC-C in the Packet Tracer workspace.
2) Cabled R1-R2, R2-R3, R1-S1, PC-A-S1, R3-S3, and PC-C-S3 using the lab port map.
3) Configured router hostnames, disabled DNS lookup, and applied interface addressing from the table.
4) Issued no shutdown on all active router interfaces and confirmed all required links came up.
5) Configured PC-A and PC-C with static IP addressing and correct default gateways.
6) Confirmed the physical topology was clean before moving into routing and security.

Routing Configuration Procedure:

1) Configured router ospf 1 on all three routers.
2) Added the 10.1.1.0/30 and 192.168.1.0/24 networks to OSPF on R1.
3) Added the 10.1.1.0/30 and 10.2.2.0/30 networks to OSPF on R2.
4) Added the 10.2.2.0/30 and 192.168.3.0/24 networks to OSPF on R3.
5) Marked G0/0/1 as passive on R1 and R3 so the LANs were advertised without forming host-facing adjacencies.
6) Validated neighbors with show ip ospf neighbor and validated learned routes with show ip route.



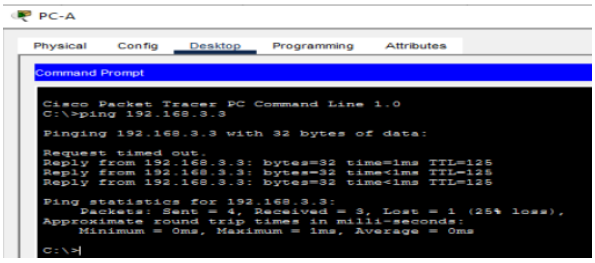
```
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.1       1   FULL/DR         00:00:31    10.1.1.1        GigabitEthernet0/0/0
192.168.3.1       1   FULL/DR         00:00:37    10.2.2.1        GigabitEthernet0/0/1
Router#
```

Figure 3. OSPF verification showing both routed adjacencies in FULL state across the three-router path.

Connectivity Validation Procedure:

1) Verified adjacent routers could see each other through OSPF before moving to end-host validation.
2) Tested router-to-router reachability from R1 to the far-side interfaces to confirm the routing plane was working.
3) Tested PC-A to PC-C end-to-end communication after the host addressing and OSPF configuration were complete.
4) Observed one initial timeout caused by ARP resolution, then repeated ping success confirmed stable end-to-end reachability.
5) Saved the working routed configuration before beginning the security hardening stages.



```
PC-A
Physical   Config   Desktop   Programming   Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=1ms TTL=125
Reply from 192.168.3.3: bytes=32 time<1ms TTL=125
Reply from 192.168.3.3: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 4. Successful end-to-end ping from PC-A to PC-C after OSPF routing and ARP resolution stabilized.

Manual Security Procedure on R1:

1) Enabled service password-encryption and set the minimum password length requirement.
2) Configured enable secret cisco12345 and the domain name netsec.com.
3) Created local users user01 and admin using secret-based syntax compatible with the Packet Tracer router image.
4) Configured the MOTD banner, console settings, AUX settings, and VTY line settings.
5) Generated RSA keys and enabled SSH version 2 support with timeout and retry controls.
6) Completed the R1 hardening phase before beginning automated security work on R3.

AutoSecure Procedure on R3:

1) Ran auto secure from privileged EXEC mode on R3.
2) Marked R3 as internet-connected and selected GigabitEthernet0/0/0 as the internet-facing interface.
3) Used the lab-provided values for the banner, enable secret, enable password, and admin account.
4) Accepted SSH configuration and default login-attack settings, but declined CBAC firewall and TCP intercept.
5) Applied the generated configuration to the running configuration and allowed AutoSecure to disable unnecessary services.
6) Verified the resulting configuration and then corrected SSH support manually where Packet Tracer required regenerated RSA keys.

KEY COMMANDS, TROUBLESHOOTING, AND EVIDENCE

Key Commands:

BASIC SETUP	ROUTING / VERIFY	SECURITY / SSH
enable	router ospf 1	service password-encryption
configure terminal	network 192.168.1.0 0.0.0.255	security passwords min-length 10
hostname R1	area 0	enable secret cisco12345
hostname R2	network 10.1.1.0 0.0.0.3 area 0	ip domain-name netsec.com
hostname R3	network 10.2.2.0 0.0.0.3 area 0	username user01 secret user01pass
no ip domain-lookup	network 192.168.3.0 0.0.0.255	username admin privilege 15
interface g0/0/0	area 0	secret adminpasswd
interface g0/0/1	passive-interface g0/0/1	banner motd " Unauthorized access
ip address 10.1.1.1	show ip ospf neighbor	is strictly prohibited! "
255.255.255.252	show ip route	line vty 0 4
ip address 10.1.1.2	ping 10.2.2.1	transport input ssh
255.255.255.252	ping 192.168.3.1	login local
ip address 10.2.2.1	ping 192.168.3.3	crypto key generate rsa
255.255.255.252	copy running-config	general-keys modulus 1024
ip address 10.2.2.2	startup-config	ip ssh version 2
255.255.255.252		auto secure
ip address 192.168.1.1		show ip ssh
255.255.255.0		show running-config | include
ip address 192.168.3.1		username
255.255.255.0		ssh -l admin 192.168.3.1
no shutdown		
show ip interface brief		

Problems Encountered and Fixes Applied:

- Packet Tracer did not accept the newer algorithm-type scrypt syntax shown in the lab sheet for R1.
- The issue was corrected by using enable secret and username ... secret syntax while preserving the same lab-provided values.
- AutoSecure accepted the configuration on R3, but SSH verification initially failed because Packet Tracer still reported SSH disabled until fresh RSA keys were generated manually.
- The issue was corrected by regenerating RSA keys, re-enabling SSH version 2, and forcing SSH on the VTY lines.

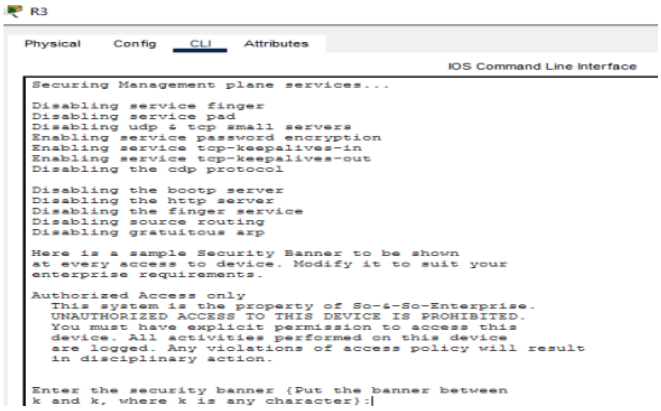


Figure 5. AutoSecure output showing multiple legacy services being disabled and the router management plane being hardened.

Evidence Collected During the Lab:

1) Topology screenshots documented the initial device placement and the final completed topology.
2) Routing screenshots documented FULL OSPF adjacency across the routed path.
3) Connectivity screenshots documented successful host-to-host communication after OSPF converged.
4) AutoSecure screenshots documented that unnecessary services and weak defaults were hardened on R3.
5) SSH screenshots documented successful remote administrative access from PC-C to R3 after the final RSA-key fix.

Operational Notes / Observations:

- Basic addressing and OSPF must be verified first because security troubleshooting is much easier on a known-good routed path.
- AutoSecure is useful because it quickly disables multiple insecure legacy services and applies protective defaults.
- Packet Tracer may require one or two extra manual SSH steps even after AutoSecure appears complete.
- End-host ping validation remains valuable because it proves the final network still works after security changes are applied.

Artifacts and Deliverables:

- Completed Packet Tracer .pkt lab file containing the fully working and hardened topology.
- Supporting screenshots retained as evidence of build, routing, AutoSecure hardening, and final SSH verification.
- Final written report documenting the completed lab workflow, security actions, and verification results.

SUMMARY, RESULTS, AND REFLECTION

Completion Verification:

- The topology was built manually and all required router, switch, and host connections were completed correctly.
- Router addressing, PC addressing, OSPF routing, passive interfaces, and end-to-end host communication were verified.
- R1 was manually hardened with encrypted secrets, local users, banner, VTY restrictions, and SSH settings.
- R3 was hardened with AutoSecure and then verified through successful SSH login from PC-C.
- The final Packet Tracer file was saved in a working, submission-ready state after all validation steps were complete.

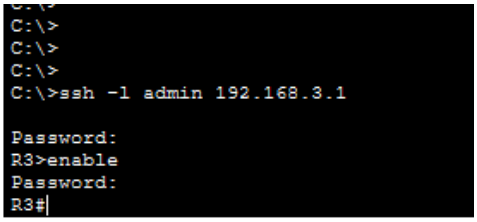


Figure 6. Final SSH verification showing successful remote administrative access from PC-C to R3.

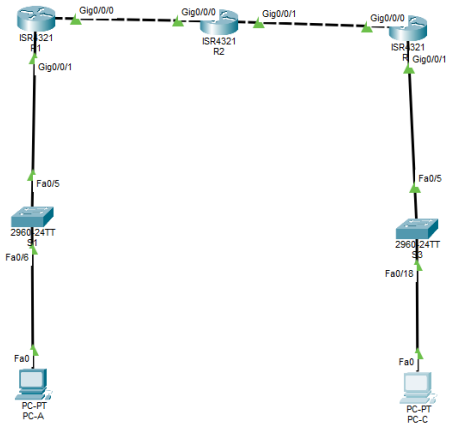


Figure 7. Final topology with the routed and secured network fully operational in its completed state.

Key Findings:

1) Reliable router hardening still depends on a correct addressing and routing baseline underneath it.
2) AutoSecure provides fast baseline protection by disabling weak services and enabling stronger management settings.
3) Manual hardening offers more control, but automated hardening is valuable for fast, repeatable baseline security.
4) Verification commands and live SSH testing are necessary because configuration text alone does not prove remote access works.

Security Concepts Reinforced:

- Administrative access is a critical attack surface, so local accounts, SSH, timeouts, and banners all matter.
- Legacy services such as finger, HTTP, CDP, and small servers increase exposure when left enabled unnecessarily.
- AAA-style line authentication and SSH reduce the risk of insecure management access compared with open VTY defaults.
- Hardening should be validated operationally so that security improvements do not break required management functions.

Best Practices / Hardening Notes:

- Complete and verify the basic routed network before beginning administrative hardening.
- Use local users, strong secrets, and SSH-only remote access for baseline router management.
- Validate every major security stage immediately with show commands and an actual login test.
- Be ready to adapt syntax slightly when Packet Tracer images differ from the exact IOS examples in the lab handout.

Reflection:

- This lab reinforced that securing a network device is not only about connectivity but also about protecting how administrators reach and manage the device.
- The most useful takeaway was seeing both approaches in one lab: manual hardening on R1 and automated hardening on R3, followed by direct SSH verification to confirm the secured path still worked.
- From a professional standpoint, the completed work demonstrates hands-on ability to build, validate, harden, troubleshoot, and verify a small routed network in a clean, methodical sequence.

Next Steps:

- Attach the completed Packet Tracer file and submit it through Canvas as required by the assignment.
- Store the final screenshots and report in the course repository as polished evidence of the completed lab.
- Reuse the same phased workflow in future labs involving router hardening, AAA, ACLs, and broader management-plane security.